

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 8: Appendix

by

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline that produced the numerics in Chapter 4: data sources, processing stages, default parameters, proxy formulas, and a glossary. The quantities cited here also appear in the tables and figures of Chapter 4.

8.1.1 Pipeline stages

The evaluation proceeds in two stages on Arbitrum One over the observation window 2025-12-01 to 2026-05-31 (UTC, inclusive).

1. Reputation graphs and scores: Extract ERC-20 Approval and Transfer logs in monthly batches (per-query dry-run and a 150 GiB abort cap). The matched cohort is wallets with non-zero approval or transfer connectivity ($n = 5,521$). Build the latest-allowance endorsement graph and the time-decayed transfer graph, then compute EndorseRank and AWP scores by power iteration (damping $d = 0.85$, tolerance 10^{-8} , at most 300 iterations); compute C-PR on the per-node mixture of the two layers at $\lambda \in \{0.25, 0.5, 0.75\}$ and S-PR on the transfer layer with the EndorseRank restart vector, under the same solver settings.
2. Checks, alignment, and benchmarks: Compute transfer and allowance checks defined in Chapter 3, report Spearman ρ and Kendall τ against each score vector with 95% paired-bootstrap intervals (400 wallet resamples, seed 42), the five pre-specified single-layer $\Delta\tau$ contrasts and the eight hybrid contrasts, time PageRank solves on pre-built edge lists under one protocol (one untimed warm-up, five timed repeats, per-session memoisation), and run sensitivity sweeps over damping, top tokens, and sample definition.
3. Temporal holdout: Freeze scores at 28 February 2026 on the spender cohort ($n = 1,335$),

count new approval pairs and new transfer senders in March–May 2026, and report every score and both raw-degree baselines against both labels with 400 paired resamples and the eight pre-registered holdout contrasts (`scripts/run_holdout_eval.py--cohortspenders`).

The tables of Chapter 4 were generated by `scripts/run_dissertation_eval.py--real--robustness--export-latex--benchmark-repeats5` and `scripts/run_holdout_eval.py--cohortspenders` on 14 September 2026, then written to `results/tables/` from the machine-readable summaries. Those summaries (`eval_summary.json`, `holdout/eval_summary_spenders.json`) and a copy of the configuration in force are committed under `A-Skill-Programs/margin_rank/results/`, so every number in the tables can be checked against the summary without re-running the pipeline. The evaluation source (`A-Skill-Programs/margin_rank`), the versioned configuration (`config/margin_config.yaml`) and the summaries are in the public repository https://github.com/abitocodes/phd_works at revision [revision-pending-commit]. The coupling weights, holdout labels and success rules were written to the configuration and to the plan of 14 September 2026 before the coupled scores were computed.

Offline synthetic checks use the same scoring and alignment logic on fixture graphs (no cloud access) to validate implementation before real extraction. Real extraction requires Google Cloud credentials and a billing project; scans target only the public Arbitrum One log dataset.

8.1.2 BigQuery extraction design

ERC-20 Approval and Transfer events are identified by their standard topic-0 hashes and restricted to logs whose owner/spender or from/to address lies in the evaluation wallet list, so scan cost scales with the cohort rather than the full chain. Monthly batching keeps each query under the operational byte cap. Representative SQL patterns appear in Appendix 8.2.

Extraction records (bytes processed, row counts, timestamps) are retained in the study’s acquisition log for auditability; this research reports the resulting cohort sizes and edge counts rather than those operational logs.

A second-tier extraction was planned for the 93,559 supplemental addresses of the expanded wallet pool ($N = 99,080$; Section 3.12.1). The dry-run estimate for the six monthly approval and transfer queries over that address set exceeded 1.3 TiB, and the download stage failed repeatedly on the storage-backed result path; the extraction was therefore abandoned within the study period. The supplemental addresses consequently enter the expanded-pool experiments as nodes only, with no approval or transfer edges of their own, which is why the full-pool row of Table 4.2 reproduces the matched-cohort graph. Completing that extraction is listed as future work in Chapter 6.

8.1.3 Scaling tables

Table 8.1 reports in-cohort runtime scaling for $n \leq 5,521$ under the timing protocol of Section 3.11; its last row is the same measurement as Table 4.1. Unlike the expanded-pool stages, every in-cohort stage is a subset of wallets whose edges were extracted, so $|V|$ and $|E|$ grow with n here.

Table 8.1: In-cohort runtime scaling (SHA256 seed benchmark-scale-v1; $n \leq 5,521$ matched GMX cohort). Here both $|V|$ and $|E|$ grow with n because every stage is a subset of wallets whose edges were extracted. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
1,000	0.056	0.502	9.0×	1,302	2,584	6,463	23,058
2,000	0.170	2.055	12.1×	2,362	5,147	11,849	47,031
4,000	0.368	3.366	9.2×	4,645	10,557	22,591	97,105
5,521	0.401	4.105	10.2×	6,442	14,727	30,791	137,087

8.1.4 Credit-delegation extract (deferred)

Aave V3 BorrowAllowanceDelegated events were not extracted in this study. Any ranking that depends on credit-delegation edges is therefore undefined on the present sample and is reported as non-interpretable ($\tau = \text{—}$) in the archived Aave-oriented matrix (Appendix 8.4). Extending the study to delegation-based reputation would require a dedicated extract of those events and a fresh alignment evaluation; it is left to future work.

8.1.5 Supplementary alignment summary table

Table 8.2 lists family-mean τ for all six proxy families computed by the pipeline, including the liquidation, inverse-risk and GMX trading-success families that were withdrawn from the primary comparison together with the GF-PR reference (Section 2.8). The transfer, allowance and Sybil-stability rows are those of Table 4.3.

Table 8.2: Mean Kendall τ by proxy family: EndorseRank and AWP ($n = 5,521$).

Family	EndorseRank	AWP	Higher
Transfer	0.333	0.534	AWP
Allowance	0.355	-0.030	EndorseRank
Liq.	0.093	0.113	AWP
Inv.risk	0.035	0.107	AWP
Sybil	0.233	0.286	AWP
GMX	0.096	0.179	AWP

8.1.6 Parameter summary

Table 8.3: Default parameters used in the primary evaluation.

Parameter	Value	Role
PageRank damping d	0.85	Default teleport mixture
Convergence tolerance ε	10^{-8}	Power-iteration stop
Max iterations	300	Power-iteration cap
Logistic k	0.01	AWP decay slope
Logistic t_0	180 days	AWP decay midpoint
Coupling weight λ	0.5 (primary); 0.25, 0.75	C-PR allowance-layer weight, fixed before the run
S-PR restart vector	normalised $\mathbf{s}_{\text{ER}}$, floor 0	Endorsement-seeded teleportation
Holdout freeze / outcome window	2026-02-28 / 2026-03-01 – 2026-05-31	Spender cohort $n = 1,335$
Observation window	2025-12-01 – 2026-05-31	UTC inclusive
Bootstrap resamples / seed	400 / 42	Paired wallet bootstrap, percentile 95% CI
Timed repeats / warm-up	5 / 1	Runtime protocol
In-cohort subsample seed	benchmark-scale-v1	Deterministic SHA256 seed
Expanded pool subsample seed	benchmark-tier2-v1	Deterministic SHA256 seed

8.1.7 Proxy operational definitions (reference)

The following formulas match Chapter 3.

- Transfer: $\text{in_degree}(w) = |\{u : \exists u \rightarrow w\}|$; $\text{in_value}(w) = \sum_{e:u \rightarrow w} \text{value}(e)$.
- Allowance: $\text{in_approve_degree}(w) = |\{o : \exists o \rightarrow w\}|$; $\text{in_approve_value}(w) = \sum_o w_a(o, w)$ with w as spender.
- Sybil-adjusted stability: $\text{inbound_counterparty_ratio}(w) = |\{u : u \rightarrow w\}|/|T^{\text{in}}(w)|$ (self-transfers excluded); tenure in days; active months.

8.1.8 Glossary

Allowance ERC-20 authorization for a spender to transfer tokens on behalf of an owner.

Alignment Rank agreement between reputation scores and a validation proxy (ρ, τ) .

Construct overlap Shared data source between a ranking method and a proxy, inflating alignment.

Endorsement graph Directed graph of latest allowances (EndorseRank input).

Matched cohort Primary sample of $n = 5,521$ wallets with non-zero endorsement- and transfer-graph connectivity.

Expanded wallet pool Larger address set ($N = 99,080$) used for node-count sensitivity of runtime; supplemental addresses have no extracted edges.

Paired bootstrap Resampling wallets with replacement using one shared index matrix for every coefficient, so that differences between coefficients can be tested on the same resamples.

PageRank Spectral ranking by stationary distribution of a damped random walk.

Proxy Observable wallet metric used for domain-intuition validation.

Teleport Uniform jump in PageRank with probability $1 - d$.

8.2 SQL Templates, Configuration, and Extended Notes

This appendix collects representative SQL patterns, configuration keys, and extended methodological notes that support reproduction.

8.2.1 Approval log extraction pattern

ERC-20 Approval events are identified by topic 0 equal to the keccak256 hash of Approval(address,address,uint256). A simplified monthly query pattern is:

```
SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS owner_topic,
    topics[SAFE_OFFSET(2)] AS spender_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
    AND block_timestamp <  TIMESTAMP('2026-01-01')
    AND topics[SAFE_OFFSET(0)] =
    ↪ '0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925'
    AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
        ↪ UNNEST(@wallet_list)
        OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
        ↪ UNNEST(@wallet_list)
    );
```

Wallet filtering via @wallet_list keeps scans proportional to the cohort rather than the full chain. Monthly batching respects per-query byte caps documented in the study’s acquisition record.

8.2.2 Transfer log extraction pattern

Transfer events use topic 0 for Transfer(address,address,uint256):

```
SELECT
    block_timestamp,
    block_number,
    transaction_hash,
    log_index,
    address AS token,
    topics[SAFE_OFFSET(1)] AS from_topic,
    topics[SAFE_OFFSET(2)] AS to_topic,
    data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
      AND block_timestamp <  TIMESTAMP('2026-01-01')
      AND topics[SAFE_OFFSET(0)] =
      ↪ '0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'
      AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
        ↪ UNNEST(@wallet_list)
        OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
        ↪ UNNEST(@wallet_list)
      );
```

Post-processing applies token decimals, builds time-decayed edge weights, and restricts nodes to the evaluation sample before PageRank.

8.2.3 Configuration keys

The pipeline’s parameters are collected in a single versioned configuration file. Salient keys include:

- observation window start and end timestamps;
- PageRank damping, tolerance, and maximum iterations;
- logistic decay k and t_0 ;
- benchmark repetition count and subsample seeds;
- output locations for intermediate artifacts and table fragments (results/tables/).

Changing damping or decay parameters requires re-running the full evaluation, including robustness sweeps and table export, and regenerating figures from the same artifacts, so that reported numerics remain consistent with the pipeline’s machine-readable summary.

8.2.4 Complexity notes

Let n be the number of nodes and $m = |E|$ the number of edges. Building latest-allowance edges is $O(R_a \log R_a)$ if R_a approval rows are sorted by timestamp, or $O(R_a)$ expected time with a hash map keyed by (token, owner, spender). Building transfer edges is $O(R_t)$ over transfer rows. PageRank is $O(m + mI)$ for I iterations, the first term being the assembly of the sparse operator. Iteration counts and edge counts for both methods are given in Table 4.1; EndorseRank needs roughly a third of AWP’s iterations and $m_{\text{approve}} \ll m_{\text{transfer}}$, so both factors favour EndorseRank.

Memory is dominated by edge lists and score vectors. The peak-memory ratio in Table 4.1 is of the same order as the edge-count ratio.

8.2.5 Conventions for the result tables

The following conventions apply to the tables of Chapter 4:

1. Identify whether a row is a social method (EndorseRank or AWP).
2. Match each family to its intended construct (allowance vs. transfer).

3. Prefer relative comparisons on the same cohort over absolute τ thresholds.
4. Read runtime tables as PageRank solve time on pre-built edge lists, not end-to-end ETL time, and expect identical numbers wherever the same graph is tabulated under the same settings (Section 3.11).
5. Read every τ with its interval, and read differences between coefficients from Table 4.4 rather than by subtracting point estimates.
6. Treat the sample-definition sweep (Table 4.10) as a sensitivity analysis whose absolute values are not comparable with the matched-cohort tables.

8.2.6 Archived extended baselines

Three-method (with the withdrawn GF-PR reference), seven-method and six-Aave alignment matrices are archived alongside the evaluation pipeline, together with corresponding CSV matrices. They are not part of the primary comparison because several variants require additional event types (e.g., credit delegation) or introduce constructs outside the EndorseRank–AWP question.

8.2.7 Limitations

SQL templates do not imply that other chains were queried, and archived baselines do not constitute peer comparisons in this research. The reported numerics are those in the pipeline’s machine-readable summary for the matched cohort ($n = 5,521$) and expanded wallet pool ($N = 99,080$).

8.3 Elementary Properties of PageRank and Rank Correlation

This appendix records elementary properties used implicitly in Chapters 2–4. The material is standard ^{[15],[20],[21],[23]}.

8.3.1 Existence and uniqueness of PageRank

Let P be a row-stochastic matrix on n nodes, possibly after dangling-node repair as in Equation (2.1). For $d \in (0, 1)$, the Google matrix

$$G_{\text{PR}} = dP^\top + \frac{1-d}{n}\mathbf{1}\mathbf{1}^\top$$

is positive and column-stochastic (up to the transpose convention in Equation (2.2)). By the Perron–Frobenius theorem for positive matrices, G_{PR} has a unique eigenvalue of modulus one with a strictly positive eigenvector π normalized so that $\sum_i \pi_i = 1$. Hence PageRank scores exist and are unique for any directed graph once damping and dangling-node rules are fixed [23].

This uniqueness underpins comparisons between EndorseRank and AWP: given fixed parameters, each method induces a single score vector on a given edge set. Differences between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ are therefore attributable to edge semantics and weights, not to solver multiplicity.

8.3.2 Continuity in edge weights

PageRank is a continuous function of the entries of P for fixed $d \in (0, 1)$. Small perturbations of allowance or transfer weights therefore induce small perturbations of scores in the $\|\cdot\|_1$ norm, with Lipschitz constants depending on d and n [23]. Continuity justifies robustness checks that vary damping or restrict to top tokens: if alignment were knife-edge fragile, small hyperparameter changes would scramble family-level τ . Empirically, EndorseRank’s allowance family mean is unchanged to three decimals across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.8), consistent with a stable ranking geometry.

8.3.3 Sparsity and per-iteration cost

A sparse matrix–vector product $P^\top \mathbf{x}$ costs $O(m)$ arithmetic operations for m nonzero edges, and assembling the sparse operator from an edge list is likewise $O(m)$. Power iteration therefore costs $O(m + mI)$ for I iterations. On the matched cohort the transfer graph has roughly 9.3 times as

many edges as the endorsement graph, so even at identical I the transfer method would be about an order of magnitude more expensive; the observed iteration counts widen the gap, and the measured wall-time ratio (Table 4.1) is of the same order as the edge ratio.

8.3.4 Properties of Spearman’s ρ

Spearman’s ρ is Pearson’s correlation applied to ranks. It is invariant to strictly increasing transformations of either variable and lies in $[-1, 1]$. Perfect monotone agreement yields $\rho = 1$; perfect reverse agreement yields $\rho = -1$. Ties require mid-rank adjustments; the evaluation pipeline uses standard tie-aware implementations. Because reputation scores and proxies are continuous and heavy-tailed, rank-based association is more appropriate than Pearson correlation on raw values; the AWP baseline paper validates its ranking with the same rank statistics^[3].

8.3.5 Properties of Kendall’s τ

Kendall’s τ equals the normalized difference between concordant and discordant pairwise orderings. It admits a probabilistic reading: for two random distinct wallets, τ is the probability their relative order agrees under both rankings, minus the probability it disagrees^[21]. A τ in the mid-0.3 range, as for EndorseRank on the allowance family (Table 4.3), means a clear but imperfect tendency for the two orderings to agree.

For large n , τ and ρ are often similar in sign and magnitude but not identical. This research reports both, following the AWP baseline^[3], and uses family-mean τ with a paired-bootstrap interval^[34] as the primary summary.

8.3.6 Construct validity note

Cronbach and Meehl^[22] distinguish criterion-related validity from construct validity. In the absence of a gold-standard default label, this research pursues a construct-oriented strategy: each method is tested against checks that operationalize its intended meaning (allowance receipt for EndorseRank; inbound transfer activity for AWP) and against a temporal holdout of later new approvals. High

within-construct τ supports interpretation. Low cross-construct τ is not automatically failure; it may indicate discriminant validity.

8.3.7 Where each primary quantity is reported

The following index gives the table in which each primary quantity is established.

Quantity	Reported in
Matched-cohort size, $ V $, $ E $, runtime, SD, memory, iterations	Table 4.1
In-cohort scaling stages	Table 8.1
Expanded-pool stages ($N = 99,080$)	Table 4.2
Family-mean τ with 95% CI; inter-method τ	Table 4.3
Paired $\Delta\tau$ contrasts	Table 4.4
Per-proxy ρ , τ , CI	Tables 4.5, 4.6, 4.7
Damping, top-token, sample-definition sweeps	Tables 4.8, 4.9, 4.10
Holdout spender cohort ($n = 1,335$)	Table 4.13
Observation window	2025-12-01 – 2026-05-31 (Table 8.3)

The quantities listed above appear in the evaluation summary and in the tables of Chapter 4.

8.4 Archived Extended Baselines

Additional PageRank variants were scored on the same matched cohort and proxy families. They are not part of the primary EndorseRank–AWP research questions.

8.4.1 Seven-method alignment matrix

Seven-method matrices remain in the evaluation archive (LP-PR, CW-AWP, LF-PR, and RiskProp-style variants). They are not drawn here. EndorseRank remains the allowance-family leader among social methods; AWP (and closely related transfer variants) lead on transfer.

The archive matrices show that:

- Methods built on transfer-like edges cluster near AWP on transfer and Sybil families.
- Methods with very small edge counts can produce unstable or degenerate rankings; their run-times may be low simply because m is tiny.
- RiskProp-style rows that mirror AWP numerically indicate shared transfer structure, not independent discovery.

8.4.2 Six-method Aave-oriented matrix

An Aave-oriented preset remains in the evaluation archive. Several lending-event graphs are sparse on Arbitrum within the study window; delegation-based rows may be degenerate when BorrowA11owanceDelegated events were not extracted (Appendix 8.1).

The Aave-oriented preset remains in the evaluation archive and is not drawn here.

These matrices motivate future work on credit-delegation edges and cross-protocol lending labels, but they do not alter Claims C1–C4.

8.4.3 Why the comparison uses two methods

The primary comparison is EndorseRank versus AWP for three reasons:

1. The novel construct is allowance-based endorsement. Comparing it to the established transfer baseline^[3] isolates the research question.
2. Additional variants that share transfer edges with AWP add rows without adding independent constructs.
3. The outcome-native GF-PR reference used in the research proposal is built from the same GMX events as the trading-success proxies, so any alignment between them is construct overlap, not validation (Section 2.8); its three-method matrix is archived alongside the seven-method and six-Aave presets.

8.4.4 Reproducing archive tables

Archive tables are regenerated by the evaluation pipeline’s table export with seven-method and six-Aave presets. CSV copies are recorded alongside the table fragments.

8.4.5 Notes on archived method identifiers

The following identifiers appear in archive tables and CSV exports.

LP-PR A transfer-graph variant with alternative loss-penalized or liquidity-oriented weighting. Empirically it tracks AWP closely on transfer families, which points to shared payment-flow structure.

CW-AWP A counterparty-weighted AWP variant that down-weights repeated transfers between the same pair. It slightly reduces transfer τ relative to AWP while remaining in the same construct family.

LF-PR A low-frequency or lightly filtered graph with very small edge counts and near-zero runtimes. Rankings are not comparable to full AWP: the sparse graph is a failed baseline.

RiskProp A risk-propagation style ranking on transfer-like edges inspired by Lin et al.^[12]. On this cohort it numerically mirrors AWP on several families, which suggests that the implemented risk signal did not diverge strongly from transfer centrality under the study’s parameterization.

LiqCall-PR / Borrow-PR / Repay-PR Aave-oriented wallet-to-wallet graphs built from liquidation calls, borrows, and repays when those events are present. On Arbitrum in the study window, edge counts are small (tens to hundreds), so alignment is weak and runtimes are negligible.

Delegation-PR Intended to use credit-delegation allowances; with zero delegation edges extracted, scores are degenerate and τ is not interpretable.

Archive rows are not peer-reviewed baselines.

8.4.6 Criteria for the two-method comparison

Chapters 4–5 report EndorseRank and AWP. The remaining variants fail the following criteria:

1. Construct independence: a baseline must implement a distinct edge semantics, not a minor reweighting of transfers.
2. Data adequacy: edge counts must be large enough for stable PageRank on the matched cohort.
3. Scope: the contribution is allowance-based endorsement, not a survey of all possible PageRank variants.

Under these criteria, EndorseRank (allowance) and AWP (transfer) form a minimal sufficient set. The seven-method and six-Aave matrices remain in the archive; they are not part of Claims C1–C4.

8.5 Claim-to-Evidence Map

This appendix maps each hypothesis, research question, and empirical claim to the specific evidence in Chapter 4.

8.5.1 H1 / RQ1: Rank divergence

- Statement: EndorseRank and AWP yield distinct wallet orderings on the same cohort.
- Evidence: inter-method Kendall τ with 95% CI (Table 4.3, last row; Section 4.5); rank scatter in Figure 4.4; score distributions in Figure 4.3.
- Interpretation: related but not interchangeable rankings; O1 criterion met (Table 5.1).

8.5.2 H2 / RQ2: Construct-specific alignment

- Statement: the two methods differ on contemporaneous allowance versus transfer checks.
- Evidence: family means with CI (Table 4.3); paired $\Delta\tau$ contrasts ii and iii (Table 4.4); per-proxy detail (Tables 4.5 and 4.6); heatmap Figure 4.2.

- Qualification: within-method contrasts i and v in Table 4.4 include zero; the claim is between-method only.

8.5.3 H3 / RQ3: Computational efficiency

- Statement: EndorseRank is faster than AWP at equal n , attributable to edge count.
- Evidence: Table 4.1 (runtime, SD, memory, iterations, $|V|$, $|E|$); Table 8.1; Table 4.2; Figure 4.1.
- Qualification: expanded-pool stages add nodes without edges (Section 4.2.1); the claim does not cover edge growth.

8.5.4 RQ4: Temporal holdout

- Statement: after scores are frozen at t1, EndorseRank predicts t2 new owner–spender approvals on inbound spenders more strongly than AWP; it does not predict them more informatively than the t1 in-approve degree.
- Evidence: Tables 4.13 and 4.14 (Section 4.8): EndorseRank minus AWP lies above zero; EndorseRank minus t1 in-approve degree lies below zero; AWP minus t1 in-degree on new transfer senders lies below zero.
- Qualification: both labels are the constructs in the future tense (new approvals, new senders), not credit or trading outcomes, and both are counts of arrivals; RQ4 replaced the proposal’s trading-success comparison (Section 2.8).

8.5.5 RQ5: Coupled operator

- Statement: a per-node convex mixture of the allowance and transfer layers (C-PR) does not rank wallets better than the appropriate single-layer method; the seeded ablation (S-PR) reproduces AWP.
- Evidence: Table 4.14 (pre-registered primary rule, both parts not met); Tables 4.11 and 4.12 (same-window position, secondary rule); runtime rows in Table 4.1.

- Qualification: fixed $\lambda \in \{0.25, 0.5, 0.75\}$, uniform teleportation, transition-level mixing, one chain; the rule and the values of λ were fixed in configuration before the run (Section 3.7).

8.5.6 Claims C1–C5

- C1: runtime evidence as in H3/RQ3; specific to EndorseRank, since C-PR costs at least as much as AWP (Table 4.1).
- C2: intended-construct checks (allowance for EndorseRank, transfer for AWP) in Table 4.3.
- C3: construct-specific winners as in H2/RQ2, tested in Table 4.4.
- C4: efficiency–alignment trade-off combining C1, C3 and the spender holdout (Tables 4.13 and 4.14), bounded by the degree baselines; robustness in Tables 4.8–4.10 and Figure 4.5.
- C5: bounded negative result on coupling, as in RQ5.

8.5.7 Explicit non-claims

This research does *not* claim:

- that AWP is obsolete;
- that EndorseRank’s allowance alignment exceeds its own transfer alignment;
- that EndorseRank or AWP predicts the future growth of its own construct better than the raw t1 degree;
- that no combination of the two edge types can beat the single-layer methods; only that the tested per-node convex mixture with fixed λ does not;
- that runtime scales favourably as the edge population grows beyond the cohort graph;
- that results generalise beyond Arbitrum One and the study window without replication;
- that on-chain scores replace collateral or legal identity systems.

8.5.8 Replication artifacts

1. Merged wallet-ranking table: scores for EndorseRank, AWP, C-PR ($\lambda \in \{0.25, 0.5, 0.75\}$) and S-PR.
2. Machine-readable holdout summary for the spender cohort, including the paired contrasts and the primary-criterion verdict (committed under `results/holdout/`).
3. Machine-readable evaluation summary: authoritative same-window numerics, per-run timings, bootstrap protocol (committed under `results/`).
4. Exported table fragments (`results/tables/`).
5. Exported vector figures (`Figures/generated/`).
6. Optional archived extended-baseline matrices.

Tables and figures are regenerated by running the evaluation pipeline on real data with robustness sweeps, exporting the tables, and regenerating figures from the same artifacts (Appendix 8.1.1).

8.6 Ethics Clearance Certificate

8.6.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.16). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate Reference: [Unisa SoC/CAES ethics clearance ref: TBD] Date of issue: [TBD] (Scanned certificate to be inserted here.)
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